

Medical Insurance Charge Prediction

Stakeholder Report

Machine Learning Analysis • 2026

Executive Summary

This report presents the findings of a machine learning programme developed to predict individual medical insurance charges. Two key business questions were investigated: what drives insurance costs, and how reliably can the model predict charges for new customers.

The analysis found that smoking status is by far the most significant driver of insurance charges, with smokers paying on average nearly four times more than non-smokers. Of the five models tested, XGBoost is recommended as the most reliable and generalisable predictor of insurance charges.



Question 1: What Drives Insurance Charges?

A correlation analysis was conducted across all available customer features to identify which factors most influence insurance charges. The heatmap analysis revealed a clear hierarchy of predictive factors.

Feature	Correlation	Plain English Meaning
Smoker Status	0.79	Strong positive link — smokers pay significantly more
Age	0.30	Moderate link — charges increase with age
BMI	0.20	Weak-moderate link — higher BMI slightly raises charges
Children	0.07	Negligible link — dependants have minimal effect
Region / Sex	~0.00	No meaningful relationship with charges

Key Finding: Smoking Status

Smoking status emerged as the single most powerful predictor, with a correlation of 0.79 — far ahead of any other variable. To put this in practical terms:

- Smokers pay an average of \$32,223 per year in insurance charges
- Non-smokers pay an average of \$8,418 per year
- This represents a 283% premium for smokers — nearly 4x the cost

Secondary Factors

Age and BMI show moderate associations with charges (0.30 and 0.20 respectively), but their predictive power is substantially weaker than smoking status. Regional and demographic factors such as sex and location showed negligible correlation and contribute little to charge variation.

Model Integrity: No Multicollinearity Detected

The heatmap also confirmed that the predictor variables are sufficiently independent of one another. This is important because when two predictors are too closely related, a model can produce unstable and misleading results. No such issue was identified here, giving greater confidence in the model's outputs.

Question 2: How Accurately Can We Predict Charges?

Five machine learning models were trained and evaluated using three standard performance metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 (how much of the variation in charges the model explains).

Understanding the Metrics

- MAE: Average prediction error in dollars — lower is better
- RMSE: Similar to MAE but penalises large errors more — lower is better
- R^2 : Proportion of charge variation explained by the model — higher is better (max 1.0)

Model	MAE (\$)	RMSE	R^2
Decision Tree ⚠️ Overfit	22.95	189,511	0.9987
XGBoost (Recommended)	584.29	1,118,211	0.9924
Random Forest	949.94	3,221,322	0.9780
XGBoost (Tuned)	2,153.20	15,751,274	0.8926
k-NN	2,467.04	17,395,534	0.8814

Why Not the Decision Tree?

While the Decision Tree shows near-perfect metrics, this is a red flag rather than a positive result. A score of $R^2 = 0.9987$ strongly indicates overfitting — the model has memorised the training data rather than learning patterns that generalise to new patients. It is unlikely to perform reliably in a real-world deployment.

Recommended Model: XGBoost

The untuned XGBoost model is recommended as the best balance of accuracy and reliability:

- R^2 of 0.9924 means it explains over 99% of variation in charges
- MAE of \$584 represents a manageable average prediction error
- As an ensemble model, XGBoost is inherently more robust and less prone to overfitting than a single decision tree
- The tuned XGBoost underperformed the default version, suggesting the tuning parameters require further refinement

Recommendations

Based on the analysis, the following actions are recommended:

- Deploy XGBoost as the primary pricing model, replacing manual estimation processes where applicable.
- Prioritise smoking status in underwriting assessments, given its outsized impact on predicted charges.
- Revisit XGBoost hyperparameter tuning to investigate why the tuned version underperformed.
- Validate model performance on an independent holdout dataset before full deployment.